

AIAC Assignment - 15.5

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B21

Task 1 – Student Attendance API

Task:

Create a RESTful API for student attendance tracking.

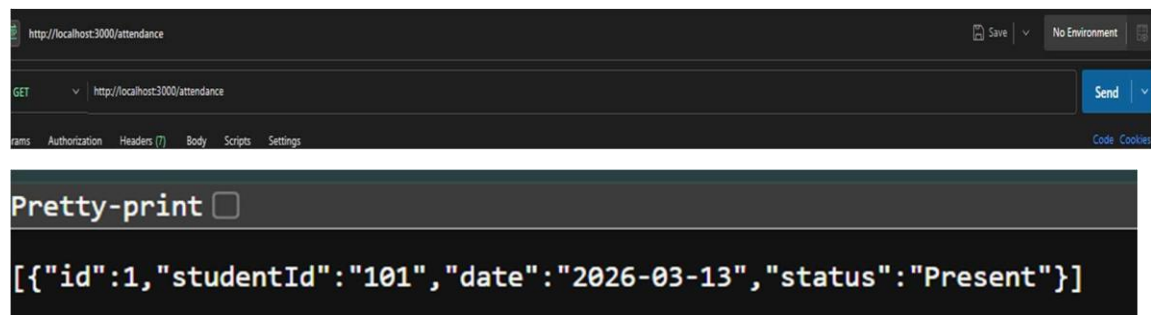
Instructions:

- Endpoints required:
 - o GET /attendance → View attendance records
 - o POST /attendance → Mark attendance
 - o PUT /attendance/{id} → Update attendance record
 - o DELETE /attendance/{id} → Remove attendance entry
- Ensure JSON-based request and response handling.

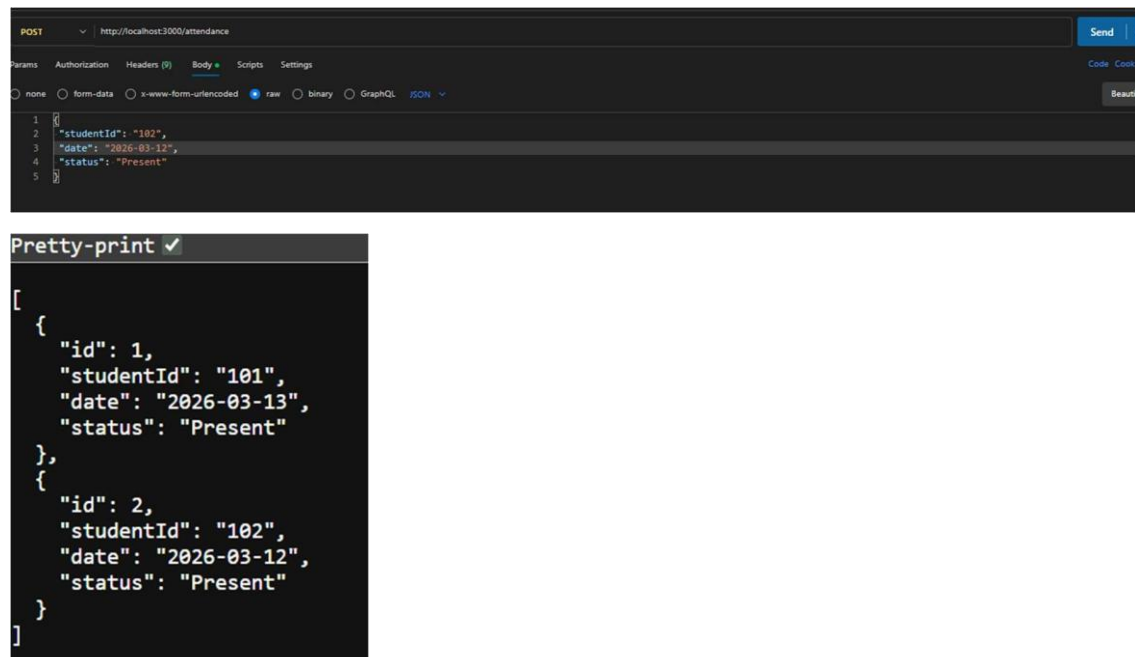
Expected Output:

- API for maintaining and updating attendance records.

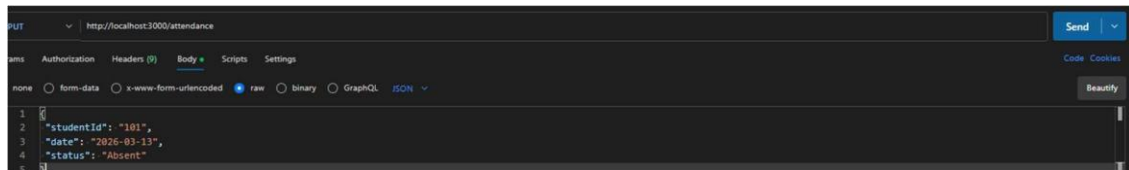
GET:



POST:

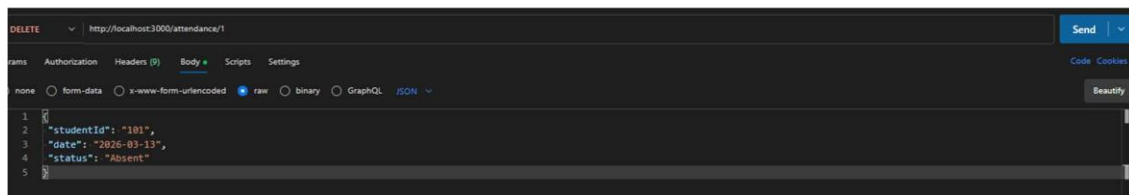


PUT:



```
[  {  "id": 1,  "studentId": "101",  "date": "2026-03-13",  "status": "Absent"  }]
```

DELETE:



```
[[  ]]
```

Task2–Inventory Management API

Task:

Develop a REST API for warehouse inventory management.

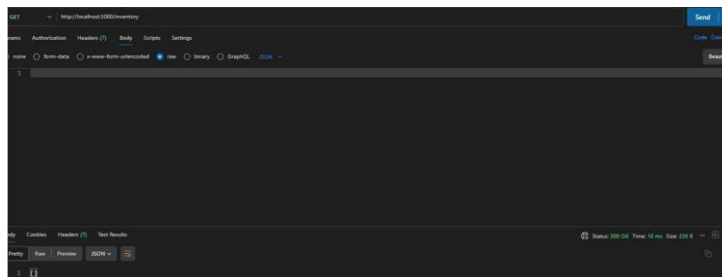
Instructions:

- Endpoints required:
 - o GET /inventory → List all items
 - o POST /inventory → Add new item
 - o PUT /inventory/{id} → Update stock quantity
 - o DELETE /inventory/{id} → Remove an item
- Implement validation for stock values.

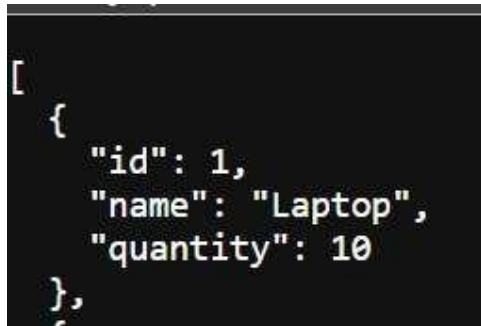
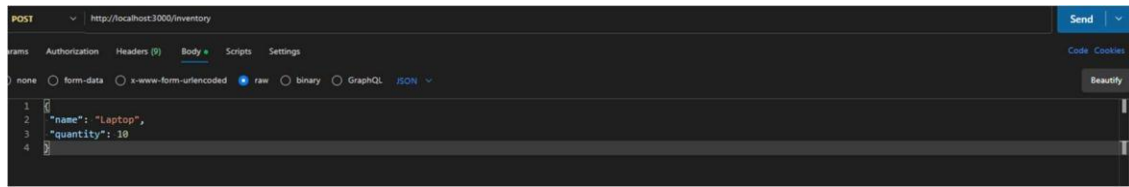
Expected Output:

- API capable of managing inventory stock levels.

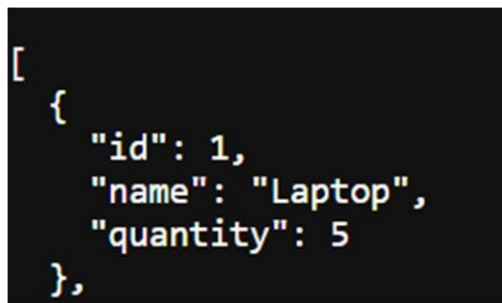
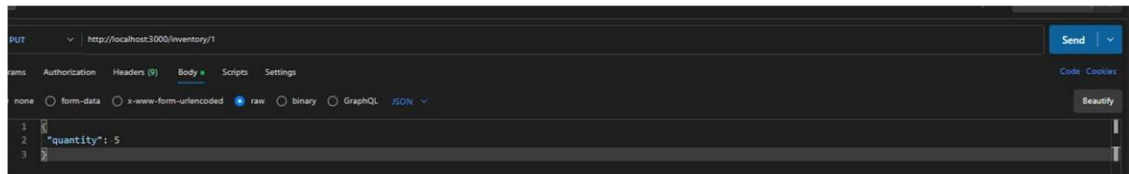
GET:



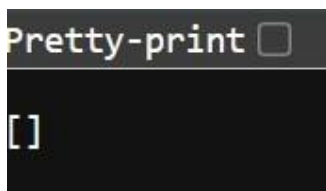
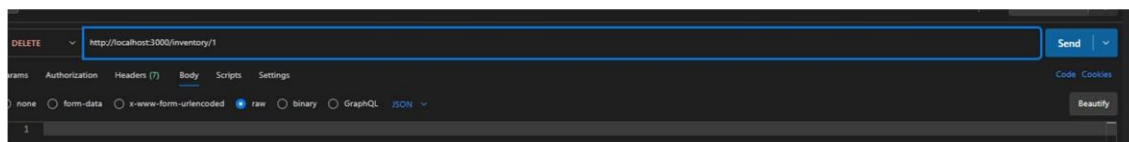
POST:



PUT:



DELETE:



Task 3 – Hotel Room Booking API

Task:

Develop a RESTful API for hotel room booking.

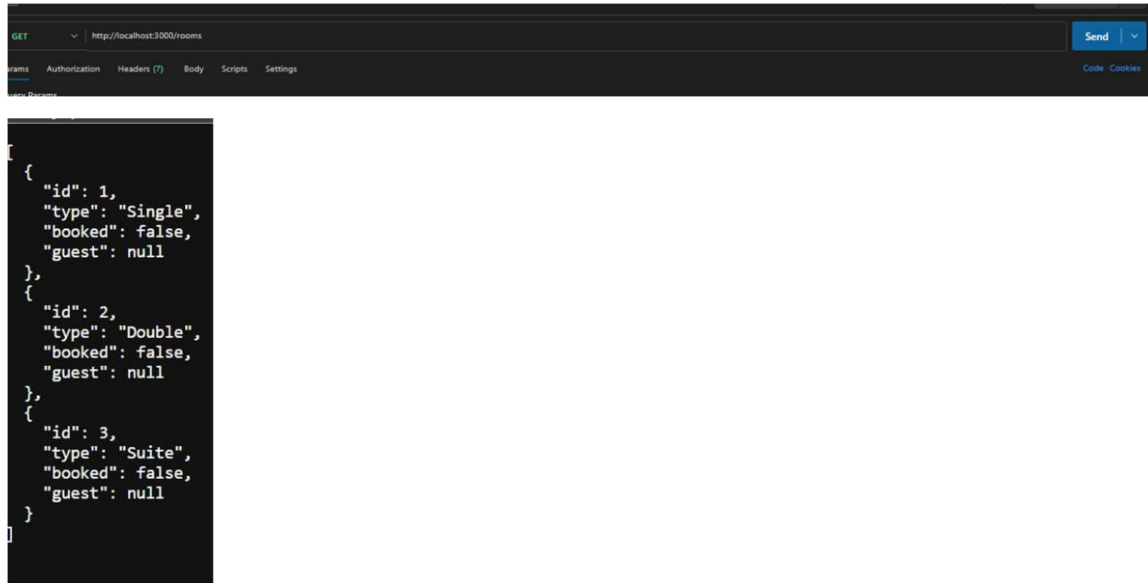
Instructions:

- Endpoints required:
 - o GET /rooms → View available rooms
 - o POST /rooms/book → Book a room
 - o GET /rooms/{id} → View booking details
 - o DELETE /rooms/{id} → Cancel booking
- Implement error handling for unavailable rooms.

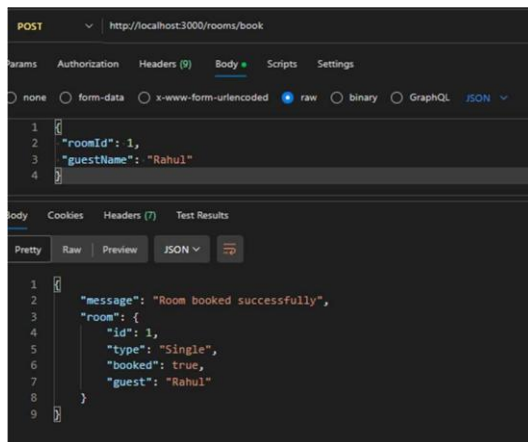
Expected Output:

- API simulating hotel room booking operations.

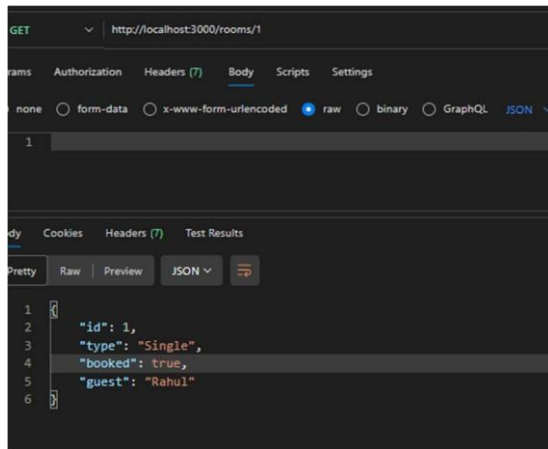
GET:



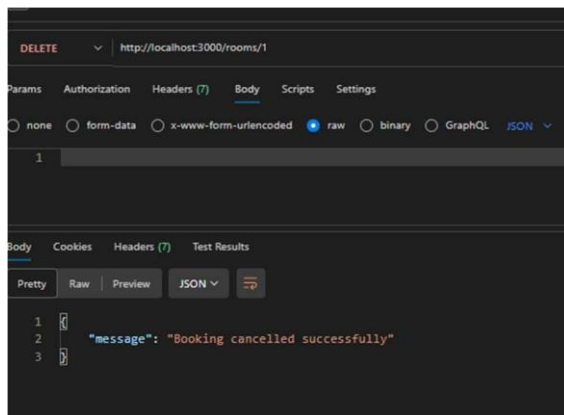
POST:



GET:



DELETE:



Task4 – Online Voting API

Task:

Create a RESTful API for an online voting system.

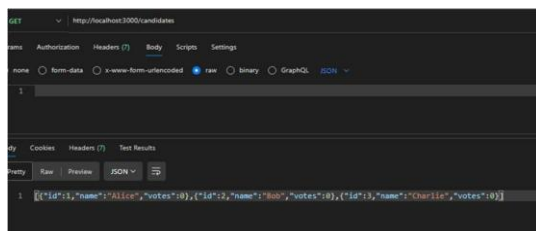
Instructions:

- Endpoints required:
 - GET /candidates → List candidates
 - POST /vote → Cast a vote
 - GET /results → View voting results
- Ensure one vote per user using in-memory validation.

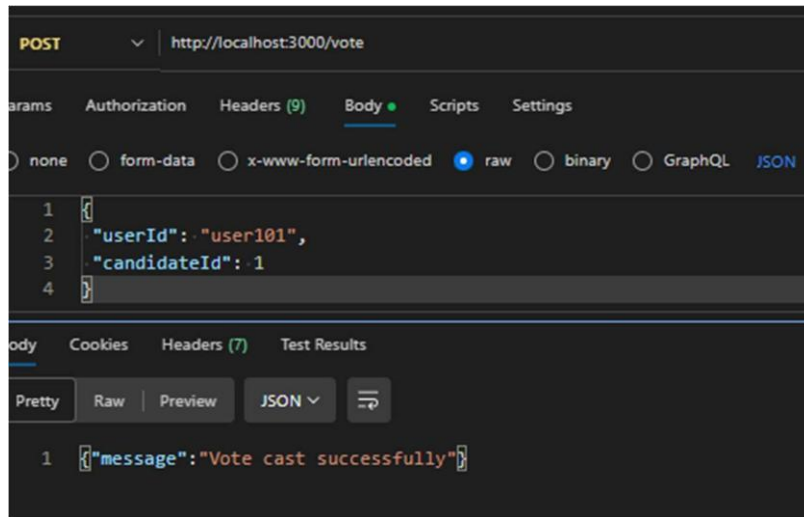
Expected Output:

- API simulating a secure online voting process.

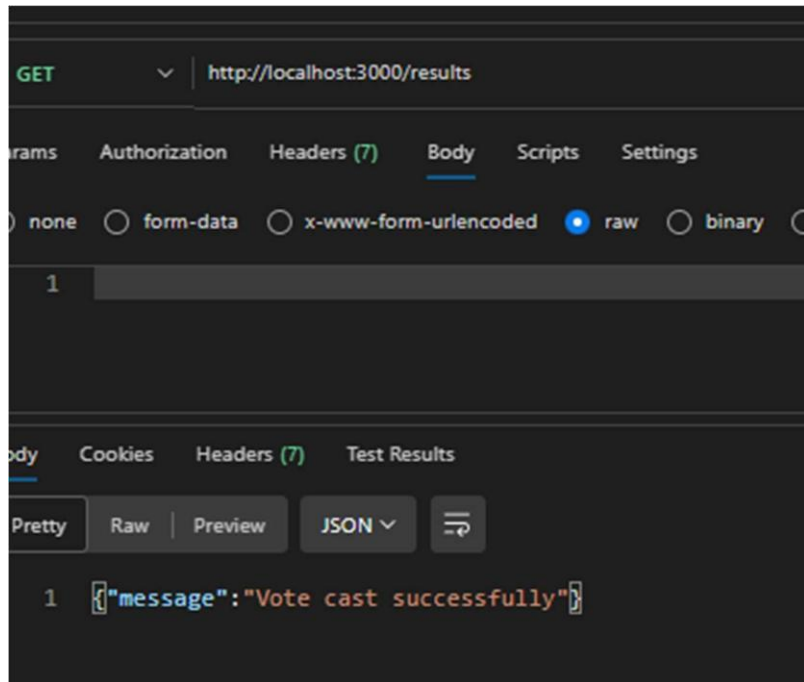
GET:



POST:



GET:



Task5 – Teacher Management API

Task:

Develop a RESTful API to manage teacher records.

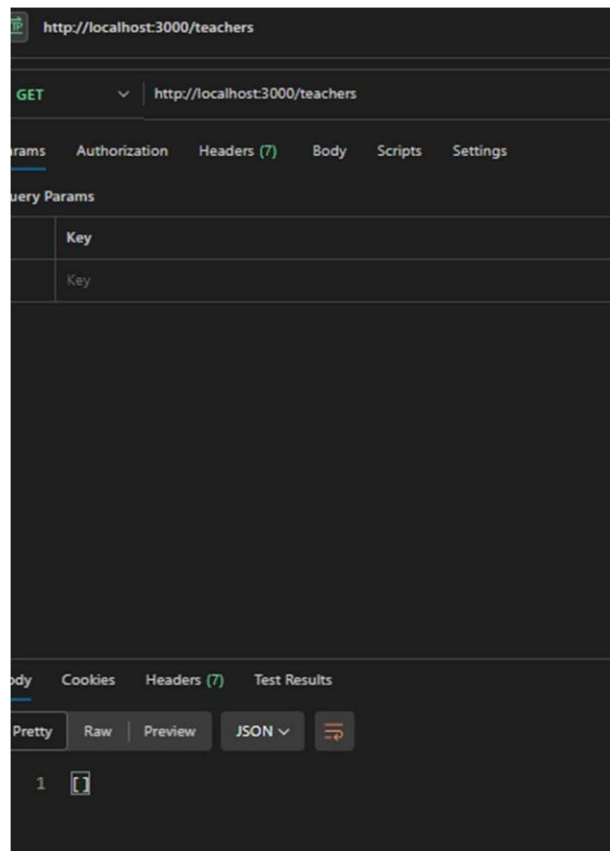
Instructions:

- Endpoints required:
 - o GET /teachers → List all teachers
 - o POST /teachers → Add a new teacher
 - o PUT /teachers/{id} → Update teacher details
 - o DELETE /teachers/{id} → Remove a teacher
- Ensure proper error handling and JSON responses.

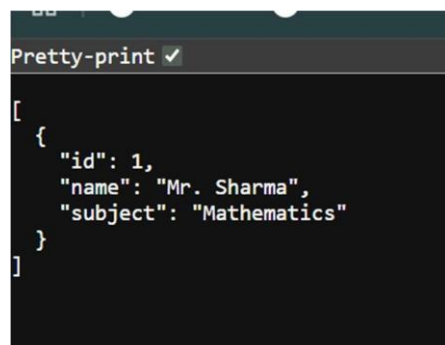
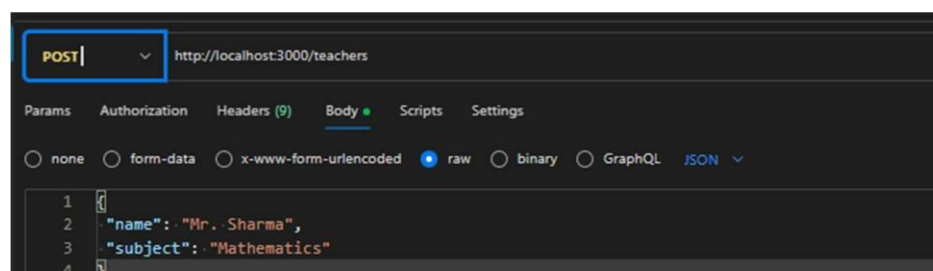
Expected Output:

- Functional API for managing teacher information.

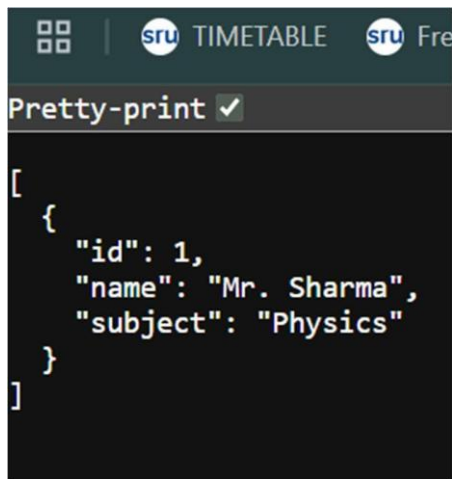
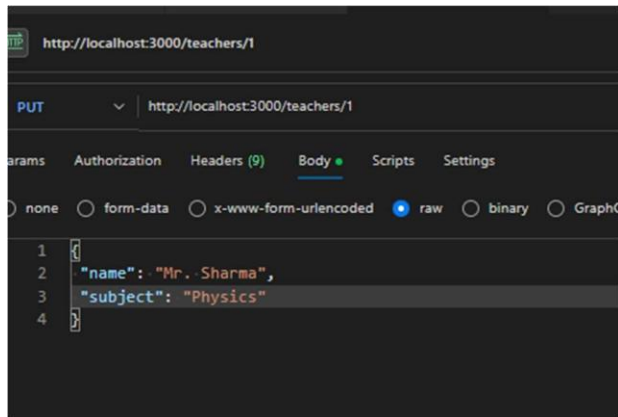
GET:



POST:



PUT:



DELETE:

